

UNIT II - SDG BASED ASSIGNMENT

Course Code: ITA0609

Course Title: Machine Learning

Student Name: S.Swagat

Explain the Role of the Error (Delta) Term in Backpropagation

Example: Neural Network Trained to Classify Malnutrition Risk in Children Under Five (SDG 2 – Zero Hunger)

Introduction

Artificial Neural Networks (ANNs) are computational models inspired by the functioning of the human brain. They consist of interconnected neurons arranged in input, hidden, and output layers. ANNs are capable of learning complex relationships from data and are widely used in healthcare, finance, agriculture, and image recognition.

One of the most popular training algorithms for ANNs is the **Backpropagation algorithm**, which adjusts the network's weights to minimise prediction errors. During training, the network repeatedly compares its predicted output with the actual target output. The difference between these two values is known as the **error**, and the **delta (δ) term** is calculated from this error. The delta term tells the network how much each neuron contributed to the error and how the weights should be modified.

In healthcare, backpropagation can be used to develop intelligent systems that predict **malnutrition risk in children under five years of age**. Such systems support doctors and healthcare workers in identifying children who require immediate nutritional care, thereby contributing to **Sustainable Development Goal (SDG 2 – Zero Hunger)**.

Understanding Backpropagation

Backpropagation is a supervised learning algorithm used to train multilayer neural networks. It works in two major phases:

1. Forward Propagation

- Input data are fed into the neural network.
- Each neuron processes the information using weights and activation functions.
- The network generates a predicted output.

2. Backward Propagation

- The predicted output is compared with the expected output.
- The error is calculated.
- The error is propagated backward through the network.
- Weights and biases are updated to reduce future errors.

This process is repeated for many training iterations (epochs) until the network achieves satisfactory accuracy.

What is the Error (Delta) Term?

The **delta (δ) term** is the error signal generated during backpropagation. It indicates how much a neuron is responsible for the prediction error.

The delta term performs three important functions:

- Measures the difference between predicted and actual output.
- Determines the direction of weight adjustment.
- Determines the magnitude of weight adjustment.

Without the delta term, the network cannot learn because it has no information about how to correct its mistakes.

Mathematical Formula

Delta for Output Layer

$$\delta = (\text{Target} - \text{Output}) \times f'(\text{Net})$$

Where

- Target = Desired output
- Output = Predicted output
- $f(\text{Net})$ = Derivative of the activation function

Delta for Hidden Layer

$$\delta_h = f(\text{Net}_h) \times \sum(\delta_o \times W)$$

Where

- δ_o = Delta of the output neuron
- W = Weight connecting hidden and output neurons

The hidden layer does not know the correct answer directly. Therefore, it learns using the error propagated back from the output layer.

Weight Update Formula

The calculated delta is used to update the connection weights.

$$w(\text{new}) = w(\text{old}) + \eta \times \delta \times x$$

Where

- η = Learning rate
- δ = Delta term
- x = Input
- w = Weight

The objective is to reduce the prediction error after every iteration.

Example: Predicting Malnutrition Risk

Suppose a healthcare centre develops a neural network to classify children into two categories:

- Healthy (0)
- Malnourished (1)

The network receives the following information:

- Age
- Weight
- Height
- MUAC (Mid-Upper Arm Circumference)
- Hemoglobin Level
- Daily Food Intake
- Number of Meals per Day
- Vaccination Status
- Recent Illness History
- Family Income

These parameters help determine whether the child is at nutritional risk.

Sample Child Data

Age = 3 years

Weight = 9 kg

Height = 82 cm

MUAC = 11.5 cm

Hemoglobin = 9 g/dL

Meals per Day = 2

Food Intake = Low

Vaccination = Complete

Recent Illness = Yes

Family Income = Low

According to medical experts, this child is malnourished.

Therefore,

Target Output = 1

Step 1: Forward Propagation

After processing all input features, the neural network predicts

Output = 0.60

This means the network estimates only a 60% probability that the child is malnourished.

Since the correct answer is 1, the prediction is not fully accurate.

Step 2: Error Calculation

Error = Target – Output

Error = 1 – 0.60

Error = 0.40

This positive error indicates that the network underestimated the child's malnutrition risk.

Step 3: Calculate Delta

Suppose the sigmoid activation function is used.

Derivative

$f'(x)$

= Output $\times$ (1 – Output)

= 0.60 $\times$ (1 – 0.60)

$$= 0.24$$

Now,

Delta

$$= \text{Error} \times \text{Derivative}$$

$$= 0.40 \times 0.24$$

$$= 0.096$$

The delta value is positive, indicating that the network should increase the output value during the next training cycle.

Step 4: Update the Weights

Assume

$$\text{Learning Rate} = 0.5$$

$$\text{Input} = 0.80$$

$$\text{Current Weight} = 0.40$$

Weight Update

$$w(\text{new})$$

$$= 0.40 + (0.5 \times 0.096 \times 0.80)$$

$$= 0.4384$$

The increased weight allows the neuron to produce a stronger activation for similar malnutrition cases in future predictions.

Hidden Layer Learning

The hidden neurons also receive an error signal.

Suppose

$$\text{Output Delta} = 0.096$$

Weight = 0.70

Derivative = 0.25

Hidden Delta

$= 0.25 \times (0.096 \times 0.70)$

$= 0.0168$

Although this value is smaller than the output delta, it enables hidden neurons to improve feature extraction.

After many training iterations, the hidden layer automatically learns important health indicators such as:

- Low body weight
- Poor growth
- Low hemoglobin
- Small MUAC
- Poor dietary diversity
- Frequent infections

Thus, hidden neurons gradually become experts in recognising malnutrition patterns.

Why is the Delta Term So Important?

The delta term is the heart of the backpropagation algorithm because it:

- Quantifies prediction error.
- Indicates whether the network output is too high or too low.
- Guides weight adjustments in the correct direction.
- Allows learning to occur in hidden layers.
- Reduces overall training loss.
- Improves prediction accuracy after every epoch.

- Prevents the network from repeating the same mistakes.
- Enables deep neural networks to solve complex healthcare problems.

Without the delta term, neural networks would simply produce outputs without improving their performance.

Advantages of Using Backpropagation for Malnutrition Detection

- Provides fast and accurate classification.
- Detects malnutrition at an early stage.
- Handles multiple health indicators simultaneously.
- Learns from thousands of patient records.
- Reduces manual diagnostic effort.
- Improves decision-making for healthcare professionals.
- Supports government nutrition programmes.
- Can be integrated with mobile health applications for rural healthcare.

Challenges

Despite its advantages, backpropagation has some limitations:

- Requires a large amount of training data.
- Training can be computationally expensive.
- Poor-quality data reduces prediction accuracy.
- May suffer from overfitting if not properly regularised.
- Requires careful selection of learning rate and network architecture.

Contribution to SDG 2 – Zero Hunger

The United Nations Sustainable Development Goal 2 aims to end hunger, achieve food security, improve nutrition, and promote sustainable agriculture.

A backpropagation-based neural network contributes to this goal by:

- Identifying children at risk before severe malnutrition develops.
- Supporting community health workers during nutritional surveys.
- Assisting hospitals in prioritising high-risk patients.
- Monitoring children's growth over time.
- Supporting government nutrition schemes such as ICDS and Poshan Abhiyaan.
- Reducing child mortality caused by severe malnutrition.
- Improving the overall health and development of children.

Thus, Artificial Intelligence becomes a valuable tool in achieving Zero Hunger through timely diagnosis and intervention.

Conclusion

The **error (delta) term** is the most important component of the backpropagation learning algorithm because it acts as the **learning signal** for the neural network. It measures the difference between the predicted and actual outputs and guides the correction of weights throughout the network. In the malnutrition risk prediction example, the delta term enables the model to gradually recognise important health indicators such as weight, height, MUAC, hemoglobin level, and food intake. With continuous training, the prediction error decreases while the accuracy increases. As a result, neural networks become reliable decision-support systems for early malnutrition detection, helping healthcare professionals provide timely treatment and significantly contributing to **SDG 2 – Zero Hunger** by improving child nutrition and reducing hunger-related health problems.